

Exercise 1: Implementing the Singleton Pattern

CODE:

```
class Logger {  
    // Single object of Logger  
    private static Logger instance;  
    // Private constructor  
    private Logger() {  
        System.out.println("Logger instance created.");  
    }  
    // Method to return the single instance  
    public static Logger getInstance() {  
        if (instance == null) {  
            instance = new Logger();  
        }  
        return instance;  
    }  
    // Logging method  
    public void logMessage(String message) {  
        System.out.println("Log: " + message);  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Logger logger1 = Logger.getInstance();  
        logger1.logMessage("Application started.");  
        Logger logger2 = Logger.getInstance();  
        logger2.logMessage("User logged in.");  
        if (logger1 == logger2) {  
            System.out.println("Only one Logger instance exists.");  
        }  
    }  
}
```

```
    } else {  
        System.out.println("Multiple Logger instances exist.");  
    }  
}  
}
```

OUTPUT:

```
Logger instance created.  
Log: Application started.  
Log: User logged in.  
Only one Logger instance exists.
```